Error and Variance Estimation for Randomized SVD
STA 243 Final Project

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Motivation and Setup

- **Paper**: Epperly, A. and Tropp, J. A. (2024). Efficient Error and Variance Estimation for Randomized Matrix Computations[2]
- **Objective**: Estimate matrix $\boldsymbol{A} \in \mathbb{R}^{d_1 \times d_2}$ by low-rank estimate $\boldsymbol{X} = g(\boldsymbol{\Omega})$, where $\boldsymbol{A}$ can only access by $\boldsymbol{\omega} \mapsto \boldsymbol{A}\boldsymbol{\omega}$ and $\boldsymbol{\Omega}$ is a random test matrix.
- **Key Challenge:** To be used safely, these algorithms require methods to assess the quality and reliability of their output.
- **Goal:** Develop diagnostics that are fast, sample-efficient (no extra matrix-vector products), and provide actionable insights for users.
- This paper introduces two main diagnostics:
  1. A **leave-one-out error estimator** ($||A - X||_F^2$) to assess the accuracy of low-rank approximations.
  2. A **matrix jackknife variance estimator** $\mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$ to measure the sensitivity of the output to algorithmic randomness.
- Our project focused on efficient diagnostics for low-rank approximations such as randomized SVD.

Prior Work: Girard-Hutchinson Norm Estimator

Formula

The estimator is based on applying random test vectors ν_i:

$$\widehat{\mathrm{Err}}^2_{\mathrm{GH}}(X, A) = \frac{1}{m} \sum_{i=1}^{m} ||(A - X)\nu_i||^2 \approx ||A - X||_F^2$$

Key Limitations

- **Costly:** Requires m additional matrix-vector products with the full matrix $\boldsymbol{A}$.
- **Less Efficient:** The proposed leave-one-out estimator is more computationally efficient and requires no new samples from $\boldsymbol{A}$.
- **Accuracy:** For a sufficient approximation rank (s), the leave-one-out method proves to be more accurate.

Leave-One-Out Error Estimator

Let $X_s = X(\omega_1, \ldots, \omega_s)$ be an approximation based on s random test vectors. Let $X^{(j)}$ be the replicate computed with vector ω_j held out.

Formula

The leave-one-out (LOO) error estimator is defined as the average squared norm of the residuals projected onto the left-out vectors:

$$\widehat{\mathrm{Err}}^2(X_{s-1}, A) := \frac{1}{s}\sum_{j=1}^{s} ||(A - X^{(j)})\omega_j||_F^2 = \sum_{j=1}^{s} Z - Q^{(j)}{Q^{(j)}}^T Z, \quad \text{where } Z = A\Omega$$

Key Property

The estimator is **unbiased** for the mean-square error of the $(s-1)$-sample approximation:

$$\mathbb{E}\left[\widehat{\mathrm{Err}}^2(X_{s-1}, A)\right] = \mathrm{MSE}(X_{s-1}, A) = \mathbb{E}||A - X_{s-1}||_F^2$$

Main Advantage

It requires **no additional matrix-vector products** with A, making it highly sample-efficient.

Related Work: Bootstrap Resampling for Variance

Estimator

Let $S = T(X)$ be a statistic of interest

1. **Resampling.** Repeat B times: draw a bootstrap matrix $\Omega^{*(j)}$ by sampling the columns $\{\omega_1, \ldots, \omega_s\}$ *with replacement*.
2. **Re-compute the estimator.** Run the same algorithm with $\Omega^{*(j)}$, producing $X^{(j)} = X(\Omega^{*(j)})$ and $S^{(j)} := T\left(X^{(j)}\right)$.
3. **Sample variance.** The bootstrap variance estimate is

$$\widehat{\mathrm{Var}}_{\mathrm{boot}}(S) = \frac{1}{B-1} \sum_{j=1}^{B} \left(S^{(j)} - \bar{S}\right)^2, \qquad \bar{S} = \frac{1}{B} \sum_{j=1}^{B} S^{(j)}.$$

Significant Limitation for Randomized SVD

- A standard bootstrap approach fails for algorithms like the Halko-Martinsson-Tropp randomized SVD.
- **Reason:** Bootstrap samples contain repeated vectors, reducing the bootstrap replicates' effective rank.
- **Result:** This leads to a gross overestimation of the true variance.

Matrix Jackknife Variance Estimator

Let X be the matrix output of a randomized algorithm. Let $X^{(j)}$ be replicates where the j-th random vector is left out, and let $X^{(\cdot)}$ be their average.

Formula

The matrix jackknife estimates the variance of the matrix output X as the sum of squared Frobenius norm deviations of the replicates from their mean:

$$\mathrm{Jack}^2(X_{s-1}) := \sum_{j=1}^{s} ||X^{(j)} - X^{(\cdot)}||_F^2$$

Key Property

The estimator **overestimates the true variance on average**, providing a conservative diagnostic:

$$\mathrm{Var}(X_{s-1}) \leq \mathbb{E}\left[\mathrm{Jack}^2(X_{s-1})\right]$$

where $\mathrm{Var}(X) := \mathbb{E}||X - \mathbb{E}X||_F^2$.

Comparison of Error Estimator

Method	Key Limitations / Failure Modes
Direct Computation	Often infeasible/too expensive for large-scale or black-box settings; requires knowing $\|A\|_F$.
GH Estimator	Cost of m additional mat–vecs with A; quality improvement not automatic with rank parameter s.
LOO Estimator	Estimates error for rank $s-1$ approximation as a proxy for rank s; accuracy depends on how well the "leave-one-out" step mimics the true rank-s error.

Table: Error-estimation Techniques

Comparison of Variance Estimator

Technique	Key Limitations / Failure Modes
Efron Bootstrap	Grossly overestimates variance for randomized SVD due to rank-reduction in bootstrap replicates; the output of Rand. SVD does not depend on column multiplicity.
Matrix Jackknife	Provides an overestimate of $\mathrm{Var}(X_{s-1})$ on average; empirical sharpness varies from dataset to dataset.

Table: Variance-estimation Techniques

Leave-One-Out Estimator

Setting:

- Let $\omega_1, \ldots, \omega_s$ be independent and identically distributed (iid) random vectors in $\mathbb{K}^{d_2}$ that are *isotropic*: $\mathbb{E}[\omega_j \omega_j^*] = \mathbf{I}$.
- Let X denote one of two matrix estimators defined for s or $s-1$ inputs:

$$X : (\mathbb{K}^{d_2})^s \to \mathbb{K}^{d_1 \times d_2} \quad \text{or} \quad X : (\mathbb{K}^{d_2})^{s-1} \to \mathbb{K}^{d_1 \times d_2}.$$

- Define estimates $X_s := X(\omega_1, \ldots, \omega_s)$, $X_{s-1} := X(\omega_1, \ldots, \omega_{s-1})$ and $X^{(j)} = X(\omega_1, \ldots, \omega_{j-1}, \omega_{j+1}, \ldots, \omega_s)$

Definition:

$$\widehat{\mathrm{Err}}^2(X_{s-1}, A) := \frac{1}{s} \sum_{j=1}^{s} \|(A - X^{(j)})\omega_j\|^2$$

Theorem 1. (Leave-one-out error estimator)

$$\mathbb{E}\widehat{\mathrm{Err}}^2(X_{s-1}, A) = \mathbb{E}\|A - X^{(j)}\|_F^2$$

Theoretical Result

Proof of Theorem 1

Note that $X^{(j)}$ be independent of ω_j.

$$\begin{aligned}\mathbb{E}_{\omega_j}\|(A - X^{(j)})\omega_j\|^2 &= \mathbb{E}_{\omega_j}\left[\omega_j^*(A - X^{(j)})^*(A - X^{(j)})\omega_j\right] \\ &= \operatorname{tr}\left((A - X^{(j)})^*(A - X^{(j)})\mathbb{E}[\omega_j\omega_j^*]\right) \\ &= \|A - X^{(j)}\|_F^2.\end{aligned}$$

By the tower property,

$$\begin{aligned}\mathbb{E}\,\widehat{\mathrm{Err}}^2(X_{s-1}, A) &= \frac{1}{s}\sum_{j=1}^{s}\mathbb{E}_{-\omega_j}\left[\mathbb{E}_{\omega_j}\|(A - X^{(j)})\omega_j\|^2\right] \\ &= \frac{1}{s}\sum_{j=1}^{s}\mathbb{E}_{-\omega_j}\|A - X^{(j)}\|_{\mathrm{F}}^2 \\ &= \frac{1}{s}\sum_{j=1}^{s}\mathrm{MSE}(X_{s-1}, A) \\ &= \mathrm{MSE}(X_{s-1}, A).\end{aligned}$$

Jackknife Variance Estimator

Setting:

- Let $\omega_1, \ldots, \omega_s$ be independent and identically distributed (iid) random elements in a measurable space $\mathcal{M}$.
- Let X denote one of two matrix estimators defined for s or $s-1$ inputs:

$$X : \mathcal{M}^s \to \mathbb{K}^{d_1 \times d_2} \quad \text{or} \quad X : \mathcal{M}^{s-1} \to \mathbb{K}^{d_1 \times d_2}.$$

- Assume X is invariant to reordering of its inputs:

$$X(\omega_1, \ldots, \omega_s) = X(\omega_{\pi(1)}, \ldots, \omega_{\pi(s)}) \quad \text{for any permutation } \pi.$$

- Define estimates:

$$X_s := X(\omega_1, \ldots, \omega_s), \quad X_{s-1} := X(\omega_1, \ldots, \omega_{s-1}).$$

Definition:

$$\mathrm{Jack}^2(X) := \sum_{j=1}^{s} \|X^{(j)} - X^{(\cdot)}\|_F^2, \quad \text{where} \quad X^{(\cdot)} = \frac{1}{s} \sum_{j=1}^{s} X^{(j)}$$

Theorem 3. (Matrix jackknife)

$$\mathrm{Var}(X_{s-1}) \leq \mathbb{E}\mathrm{Jack}^2(X)$$

Proposition 2 (Efron–Stein–Steele Inequality) [1][4]

Let $\omega_1, \ldots, \omega_s \in \mathcal{M}$ be independent random variables in a measurable space $\mathcal{M}$, and let $f : \mathcal{M}^s \to \mathbb{K}$ be measurable. Let ω_i' be an independent copy of ω_i. Then:

$$\mathrm{Var}(f(\omega_1, \ldots, \omega_s)) \leq \frac{1}{2} \sum_{i=1}^{s} \mathbb{E}\left[\left(f(\omega_1, \ldots, \omega_s) - f(\omega_1, \ldots, \omega_{i-1}, \omega_i', \omega_{i+1}, \ldots, \omega_s)\right)^2\right]$$

Proof Sketch.

- Let $Z := f(\omega_1, \ldots, \omega_s)$, $Z_i := f(\omega_1, \ldots, \omega_{i-1}, \omega_i', \omega_{i+1}, \ldots, \omega_s)$.
- Then claim is $\mathrm{Var}(Z) \leq \frac{1}{2} \sum_{i=1}^{s} \mathbb{E}(Z - Z_i)^2$. WLOG $\mathbb{E}Z = 0$
- If $\mathbb{E}Z^2 = \infty$, by telescoping argument:

$$\mathrm{Var}(Z) = \frac{1}{2}\mathbb{E}[(Z - Z_s')^2] = \frac{1}{2}\mathbb{E}\left[\sum_{i=1}^{s}(Z_{i-1}' - Z_i')\right]^2 \leq \frac{s}{2}\mathbb{E}\sum_{i=1}^{s}(Z_{i-1}' - Z_i')^2.$$

  where $Z_i' = f(\omega_1', \ldots, \omega_{i-1}', \omega_i', \omega_{i+1}, \ldots, \omega_s)$
- If $\mathbb{E}Z^2 < \infty$, expand $Z = \sum_{\mathbf{k}} c(\mathbf{k})\phi_k$ in an orthonormal basis of $L^2(dF)$.

$$\frac{1}{2}\mathbb{E}\sum_{i=1}^{s}(Z - Z_i)^2 = s\mathbb{E}Z^2 - \sum_{i=1}^{s}\mathbb{E}ZZ_i = s\mathbb{E}Z^2 - \sum_{\mathbf{k}} c^2(\mathbf{k})\#(k_i = 0) \geq \mathrm{Var}(Z)$$

Proof Sketch of Theorem 3

Apply Prop. 2 to matrix-valued function $X_{s-1} \in \mathbb{K}^{d_1 \times d_2}$.
Fix a pair of indices $1 \leq m \leq d_1$ and $1 \leq n \leq d_2$. Since the samples $\omega_1, \ldots, \omega_{s-1}$ are identically distributed and X invariant to reordering, the last sample ω_s can be used to fill the role of each ω_j'. For the (m, n)-matrix entry $(X_{s-1})_{mn}$, we observe

$$\begin{aligned}
\mathrm{Var}((X_{s-1})_{mn}) &\leq \frac{1}{2} \sum_{i=1}^{s-1} \mathbb{E}\left((X^{(s)})_{mn} - (X^{(i)})_{mn}\right)^2 \\
&= \frac{1}{2s} \sum_{i,j=1}^{s} \mathbb{E}\left((X^{(i)})_{mn} - (X^{(j)})_{mn}\right)^2 \\
&= \sum_{j=1}^{s} \mathbb{E}\left((X^{(j)})_{mn} - (X^{(\cdot)})_{mn}\right)^2 .
\end{aligned}$$

Sum this equation over all $1 \leq m \leq d_1$ and $1 \leq n \leq d_2$.

Simulation Design and Results

Randomized SVD

Randomized SVD (Halko[3]): Project A onto a low-dimensional random subspace and compute SVD in that subspace to approximate the dominant components.

Goal: Efficiently approximate the top-s singular values, vectors and the approximation $X(\Omega)$ of a large matrix $A \in \mathbb{R}^{n \times d}$.

Algorithm Randomized SVD (Sketched parameter $= s$)

Require: Matrix $A \in \mathbb{R}^{d_1 \times d_2}$, sketched parameter s, power iterations q
Ensure: Approximate SVD: $X(\Omega) := U_s \Sigma_s V_s^T \approx A$

```
Generate a Random Sketching Matrix Ω ∈ R^{d2 × s}
Y ← AΩ
for i = 1 to q do
    Y ← (AA^T)Y                                  ▷ Optional power iteration
end for
Compute Economy QR decomposition: Y = QR
C ← Q^T A
Compute SVD: C = W Σ_s V_s^T
U_s ← QW
return U_s, Σ_s, V_s
```

Randomized SVD

- Randomized SVD can significantly improve the calculation speed with relative controlled error.
- We want to design a simulation for different matrics in RSVD.

Matrix A (7952x5304)

$U\Sigma V^T$ from SVD (time: 39.83s)

$U_s\Sigma_s V_s^T$ from RSVD (time: 0.63s)

Simulation Design and Results

Simulation Design:

- Construct following three matrics for RSVD (Joel[5]);

$$\text{(NoisyLR)} \quad \mathbf{A} = \mathrm{diag}(\underbrace{1, \ldots, 1}_{R \text{ times}}, 0, \ldots, 0) + \xi d^{-1} \mathbf{G}\mathbf{G}^* \in \mathbb{R}^{d \times d}.$$

$$\text{(ExpDecay)} \quad \mathbf{A} = \mathrm{diag}(\underbrace{1, \ldots, 1}_{R \text{ times}}, 10^{-q}, 10^{-2q}, \ldots, 10^{-(d-R)q}) \in \mathbb{R}^{d \times d}.$$

$$\text{(PolyDecay)} \quad \mathbf{A} = \mathrm{diag}(\underbrace{1, \ldots, 1}_{R \text{ times}}, 2^{-p}, 3^{-p}, \ldots, (d-R+1)^{-p}) \in \mathbb{R}^{d \times d}.$$

  where $U_s \in \mathbb{R}^{n \times s}$, $\Sigma_s \in \mathbb{R}^{s \times s}$, and $V_s \in \mathbb{R}^{d \times s}$.
- In the following numerical simulation, we set $d = 1000$ and PolyDecay ($R = 5$, $p = 2$), NoisyLR ($R = 5$, $\xi = 10^{-4}$), ExpDecay ($R = 5$, $q = 0.1$).
- Our simulation is conducted in the Real Space $\mathbb{R}^d$.

Simulation Design and Results

Simulation Design:

- Denote the approximation of A by $X := U_s \Sigma_s V_s^T$. The following error diagnostics are defined with respect to this approximation.
- Use **Girard-Hutchinson** estimator and **Leave-one-out** Method to estimate the relative error:

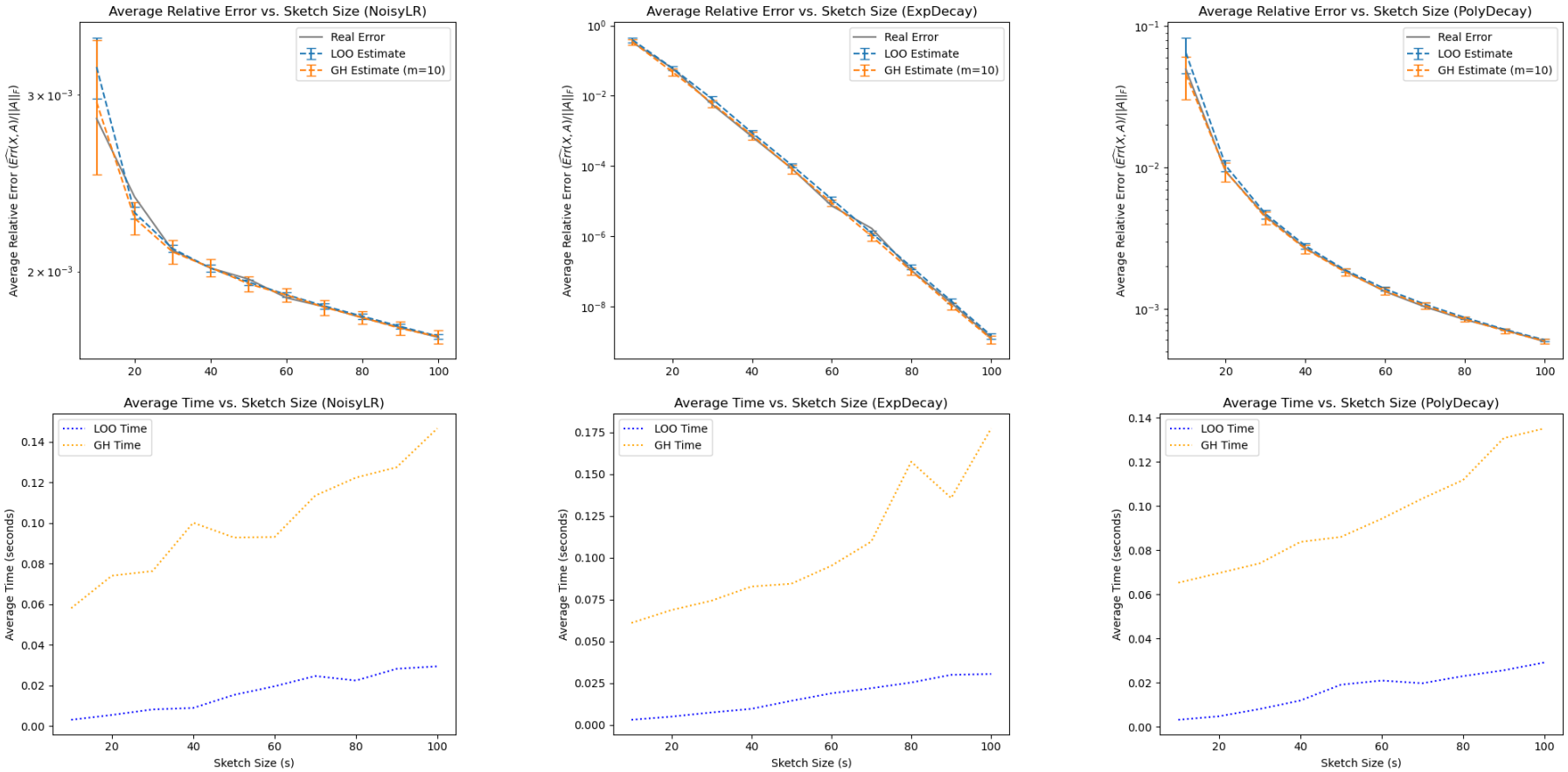
$$\text{Relative Error} = \frac{\mathrm{Err}(X, A)}{||A||_F} = \frac{||A - X||_F}{||A||_F};$$

- Use **Efron Bootstrap** estimator and **Matrix Jackknife** estimator to estimate the variance:

$$\mathrm{Var}(X) = \mathbb{E}||X - \mathbb{E}X||_F^2.$$

Simulation Design and Results

Error estimation results:

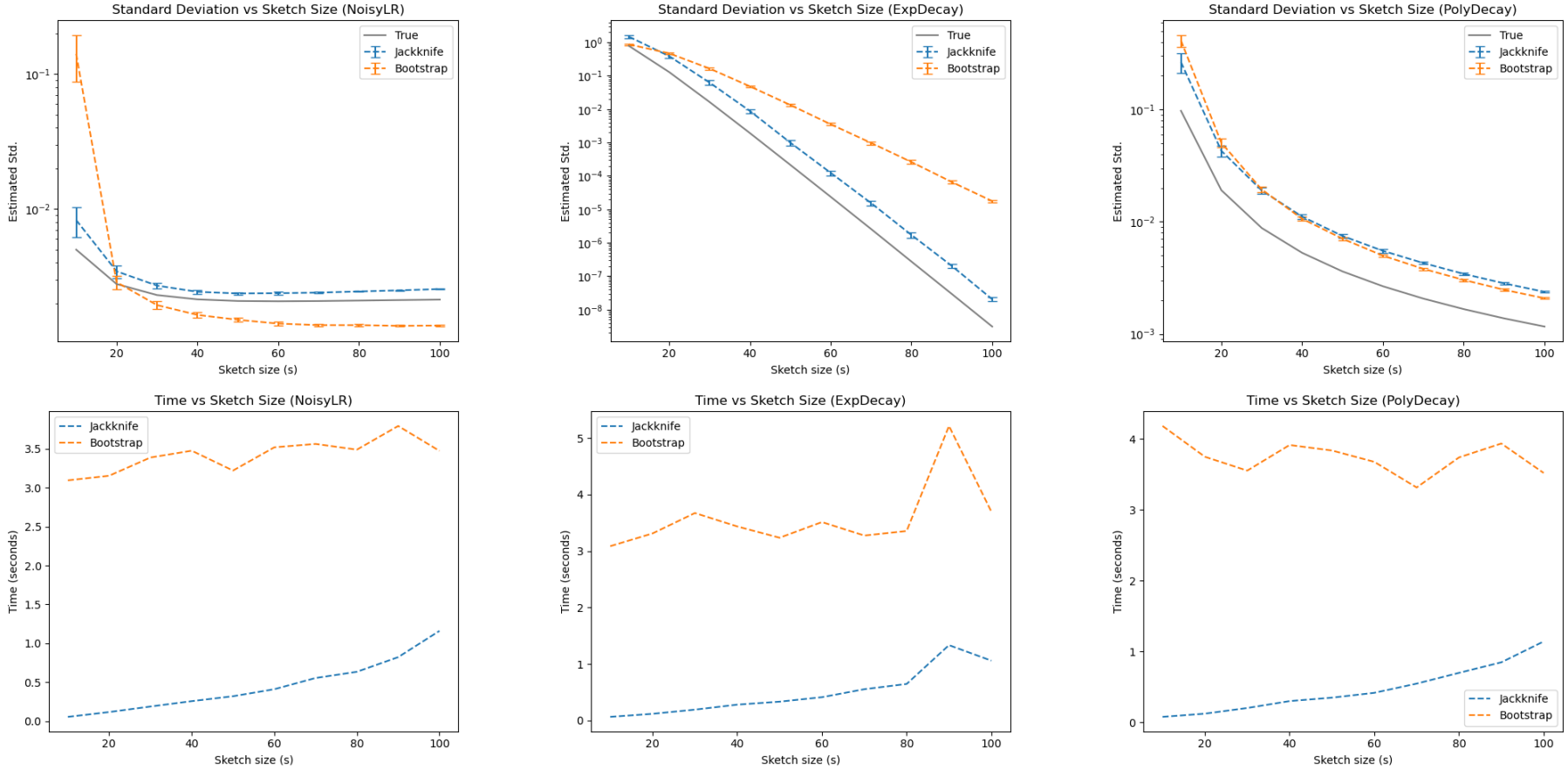


Plots Notes: rep $= 100$; $m = 10$ in GH.

- Both LOO and GH yield unbiased error estimates.
- LOO shows lower standard error.
- GH requires 5–15$\times$ more computation time than LOO.

Simulation Design and Results

Variance estimation results:



Plots Notes: rep=100; $B = 100$ in Bootstrap.

- Jack(X) provides a stable and conservative estimate of standard deviation.
- Boot(X) shows high variability—overestimating or underestimating depending on the target matrix structure.
- Bootstrap also requires 3–10$\times$ more computation time than Jackknife.

Simulation Design and Results

Matrix Type	True Std.	Estimated Std.		Ratio (Estimate / True)	
		Jackknife	Bootstrap	Jackknife	Bootstrap
PolyDecay ($R = 5$, $p = 2$)	1.16×10^{-3}	2.40×10^{-3}	2.05×10^{-3}	2.06	1.77
NoisyLR ($R = 5$, $\xi = 10^{-4}$)	2.13×10^{-3}	2.52×10^{-3}	$\mathbf{1.37 \times 10^{-3}}$	1.19	**0.64**
ExpDecay ($R = 5$, $q = 0.1$)	3.04×10^{-9}	2.28×10^{-8}	$\mathbf{2.19 \times 10^{-5}}$	7.50	**7210.00**

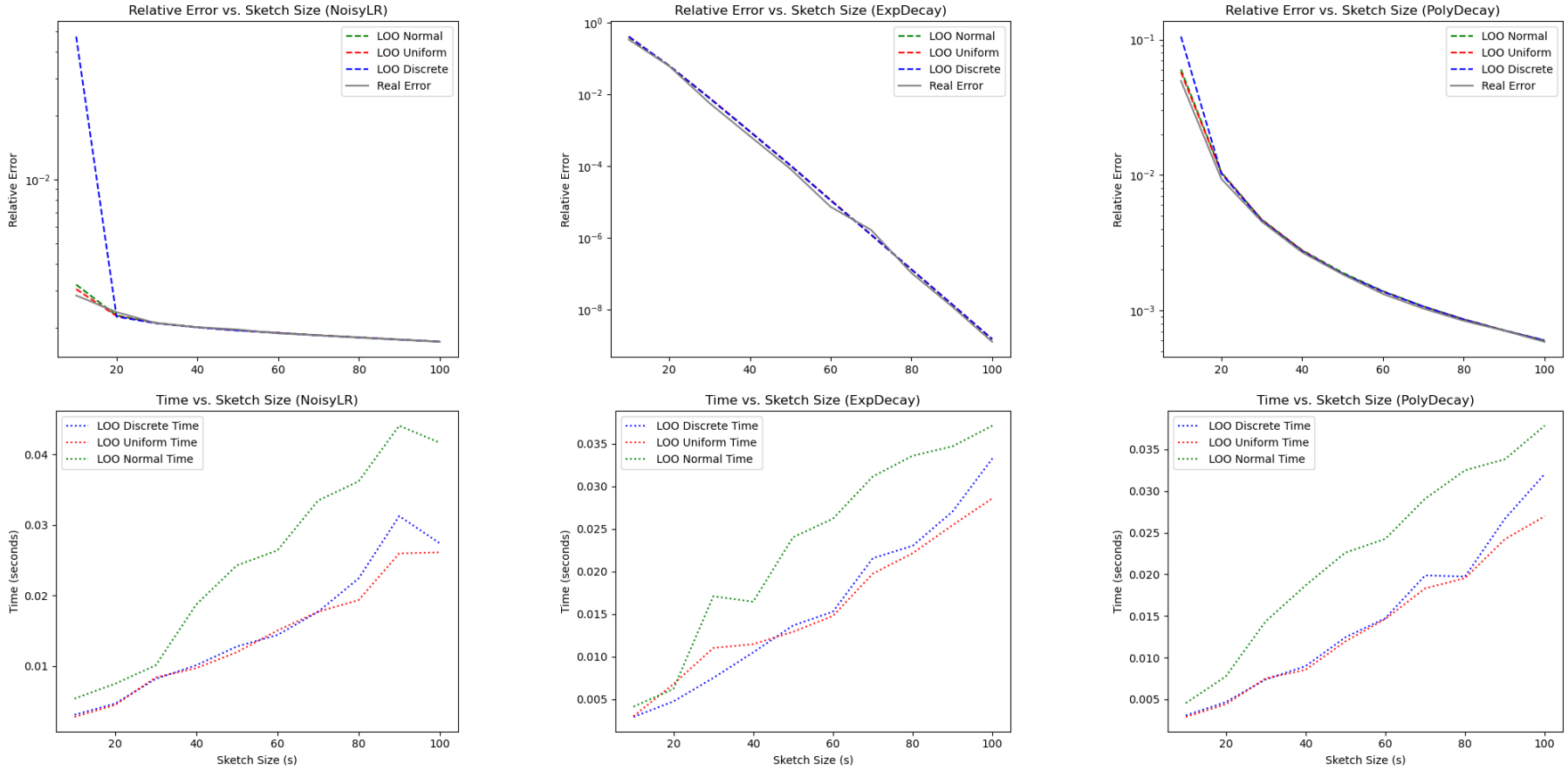
Table: Comparison of true and estimated std. and their ratios at sketch size $s = 100$

- Jackknife provides stable estimates with errors typically within $10\times$ (one order of magnitude).
- Bootstrap is unstable: it can overestimate by $10^3\times$ or underestimate by $0.5\times$.

Simulation Design and Results

Additional Findings in the choose of Ω: Random Sketching Matrix $\Omega = \{\omega_{ij}\}_{d_2 \times s} \in \mathbb{R}^2$ is generated from the following distribution:

- **Gaussian:** Each entry $\omega_{ij} \sim \mathcal{N}(0, 1)$
- **Rademacher:** $\omega_{ij} = \pm 1$ with equal probability
- **Uniform:** $\omega_{ij} \sim \mathcal{U}(-\sqrt{3}, \sqrt{3})$ to ensure variance 1



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Denote the orthonormal basis of the Hilbert space $L^2(dF)$ as $(\phi_k, k = 0, 1, \dots)$, $\phi_0 \equiv 1$. Let $\mathbf{k} = (k_1, \dots, k_s)$ and define $\phi_{\mathbf{k}}(\boldsymbol{\omega}) = \prod_{i=1}^{s} \phi_{k_i}(\omega_i)$. Then there exists constants $c(\mathbf{k})$ such that $Z = f(\omega_1, \dots \omega_s) = \sum_{\mathbf{k}} c(\mathbf{k})\phi_{\mathbf{k}}(\boldsymbol{\omega})$ Then,

$$\mathbb{E}ZZ_i = \sum_{\mathbf{k},\mathbf{l}} c(\mathbf{k})c(\mathbf{l})\mathbb{E}\left[\phi_{k_i}(\omega_i)\phi_{l_i}(\omega_i')\prod_{j\neq i}\phi_{k_j}(\omega_j)\phi_{l_j}(\omega_j)\right] = \sum_{\mathbf{k}:k_i=0} c(\mathbf{k})^2$$

Last equation is from orthogonality of ϕ_ks, independence of ω_js and ω_i', and the fact that only ϕ_0 is not centered among ϕ_ks.

$$\begin{aligned}
\frac{1}{2}\mathbb{E}\sum_{i=1}^{s}(Z - Z_i)^2 &= s\mathbb{E}Z^2 - \sum_{i=1}^{s}\mathbb{E}ZZ_i \\
&= s\mathbb{E}Z^2 - \sum_{\mathbf{k}} c^2(\mathbf{k})\#(k_i = 0) \\
&\geq s\mathbb{E}Z^2 - (s-1)\sum_{\mathbf{k}} c^2(\mathbf{k}) \\
&= s\mathbb{E}Z^2 - (s-1)\mathbb{E}Z^2 \\
&= \mathrm{Var}(Z)
\end{aligned}$$

The third line used the fact if $k_i = 0$ for all i, then $c(\mathbf{0}) = 0$ so that enough to think the number of nonzero k_i is at most $s - 1$, the fourth line is property of the basis.